

Period-Aware Inductive Bias Versus Scale on Influenza-like Illness Forecasting

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Abstract

Forecasting influenza-like illness (ILI) is important for public-health planning, but weekly surveillance series combine short-lag persistence with annual recurrence and season-to-season shifts in peak timing and amplitude. Foundation models (FMs) and large language model (LLM)-adapted forecasters are increasingly used for time-series forecasting, yet on the standard ILI benchmark they do not show a clear average-error advantage over strong compact baselines. We therefore propose EPIC (Efficient Periodic Inductive Convolution), a compact attention-free forecaster that combines temporal convolution, period projection, and latent horizon projection. Under the same benchmark setting, EPIC achieves lower four-horizon average error than published compact and LLM-based comparators, and ablations show that the temporal and periodic branches contribute complementary information. This advantage is concentrated in the average and shorter-to-medium horizons, while the longest horizon remains a limitation. For weekly ILI surveillance, a small period-aware inductive bias is more effective than additional model scale.

1. Introduction

Forecasting influenza-like illness (ILI) informs vaccine planning, hospital preparedness, and public-health response (Biggerstaff et al., 2016; McGowan et al., 2019; Reich et al., 2019; Lutz et al., 2019). Weekly surveillance series combine short-lag persistence, annual recurrence, drifting peak timing and amplitude, and coupled multivariate channels. This regime is better described as structured periodic non-stationarity than as stationary seasonality (Brooks et al., 2015). A general-purpose mixer must recover both effects implicitly from a short training history, whereas a model

that encodes them separately can exploit each as a distinct inductive bias. Which choice is more effective on weekly ILI is the question we address.

Figure 1 shows that the ILI target has strong short-lag persistence, recurring annual power with drifting peak timing, and only a short training history. The first property favors causal temporal convolutions with a small dilated receptive field; the second favors spectral period features, because the magnitude spectrum is insensitive to global phase shifts within a window; the third makes explicit periodic and temporal components more data-efficient than a single general-purpose mixer. These observations motivate **EPIC** (Efficient Periodic Inductive Convolution).

Our work makes three main contributions.

- We introduce EPIC, an attention-free ILI forecaster that combines a causal temporal-convolution backbone, an explicit period branch, and a latent horizon projector.
- We show that EPIC reaches lower four-horizon average error than published compact and LLM-based forecasters on the `national_illness` benchmark, while the same published results show no consistent average-error advantage for LLM-based comparators over compact baselines.
- We provide branch and projector ablations showing that the proposed combination is best on most individual horizons and on the average.

2. Related Work

Recent work on time-series foundation models (FMs) and LLM-based forecasting has emphasized reusable pretrained backbones and cross-domain coverage (Liang et al., 2024; Jin et al., 2024; Liu et al., 2024; 2025). In parallel, simple linear and patch-based architectures have repeatedly proved hard to beat on standard long-horizon benchmarks (Zeng et al., 2023; Nie et al., 2023), raising the question of whether scale alone is decisive in this regime.

On ILI and long-horizon forecasting benchmarks, representative compact supervised baselines include PatchTST (Nie et al., 2023), WhiteTST (Shen et al., 2024), xLSTTime (Alharthi & Mahmood, 2024), and MSCformer (Li et al.,

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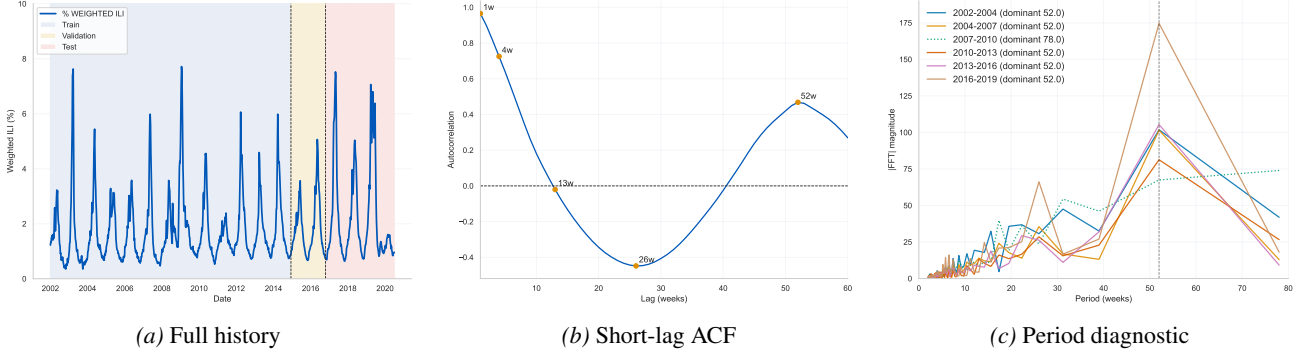


Figure 1. ILI diagnostic views on `national_illness`. The target combines recurring seasonal structure, short-lag persistence, and annual periodic power that remains visible across most non-overlapping three-year windows.

2024), while representative pretrained or LLM-based comparators include OFA (Zhou et al., 2023), Time-LLM (Jin et al., 2024), UniTime (Liu et al., 2024), TimeCMA (Liu et al., 2025), and MIFlu (Moon et al., 2025). We use these two groups as the comparison set in Section 4.

3. Methodology

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]^\top \in \mathbb{R}^{T \times C}$ denote a multivariate influenza-like illness (ILI) history with C channels and look-back window T . The task is to predict the next H steps, $\hat{\mathbf{Y}} \in \mathbb{R}^{H \times C}$. We describe the four components of EPIC in turn: input encoding with horizon projection, causal temporal convolution, a learnable period branch, and additive fusion with a linear readout.

3.1. Input Encoding and Horizon Projection

We first normalize each channel of the input sequence using instance statistics,

$$\tilde{\mathbf{X}} = \frac{\mathbf{X} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad (1)$$

where $\boldsymbol{\mu}, \boldsymbol{\sigma} \in \mathbb{R}^{1 \times C}$ are computed over the temporal dimension. The normalized sequence is embedded into a latent representation

$$\mathbf{Z}^{(0)} = \mathcal{E}(\tilde{\mathbf{X}}), \quad \mathbf{Z}^{(0)} \in \mathbb{R}^{T \times d}. \quad (2)$$

Before applying the forecasting blocks, EPIC expands the latent sequence from length T to length $T + H$ through independent latent-dimension projectors. For each latent coordinate $k \in \{1, \dots, d\}$,

$$\bar{\mathbf{z}}_{:,k}^{(0)} = \mathcal{P}_k(\mathbf{z}_{:,k}^{(0)}), \quad \mathcal{P}_k: \mathbb{R}^T \rightarrow \mathbb{R}^{T+H}. \quad (3)$$

This operation produces $\bar{\mathbf{Z}}^{(0)} \in \mathbb{R}^{(T+H) \times d}$. Projecting to $T+H$ lets all forecast positions be processed in one forward pass and avoids autoregressive error accumulation, which matters when H exceeds T .

3.2. Temporal Convolutional Interaction

EPIC stacks L identical forecasting blocks. Given the block input $\bar{\mathbf{Z}}^{(\ell-1)}$, the first stage applies a causal temporal-convolution interaction module,

$$\mathbf{U}^{(\ell)} = \text{LN}\left(\mathcal{F}_{\text{TCN}}\left(\bar{\mathbf{Z}}^{(\ell-1)}\right)\right), \quad (4)$$

where $\mathcal{F}_{\text{TCN}}$ is a residual temporal convolutional network with three residual blocks, two kernel-3 dilated convolutions per block, and dilations $\{1, 2, 4\}$. The convolution is causal over the projected latent sequence of length $T + H$, which covers short- and medium-range temporal dependence within a single forward pass. Because each forecast position is a learned function of the observed window through the projector $\mathcal{P}_k$, the model is non-autoregressive in observed time.

We use fixed local structure instead of a temporal self-attention operator, and the temporal feature is

$$\mathbf{H}_T^{(\ell)} = \mathbf{U}^{(\ell)}. \quad (5)$$

3.3. Learnable Period Branch

ILI sequences recur annually, but their peaks are not phase-locked across years. To capture this component without assuming a fixed seasonal template, EPIC augments the temporal feature with a learnable period branch. Given $\mathbf{U}^{(\ell)}$, we first compute its magnitude spectrum,

$$\mathbf{S}^{(\ell)} = \left| \text{FFT}\left(\mathbf{U}^{(\ell)}\right) \right|. \quad (6)$$

The magnitude spectrum is invariant to global shifts of a windowed signal, so it emphasizes recurring periodic content while suppressing exact within-window phase. The learnable period prototypes operate on this representation, allowing the period branch to track annual structure even when peak timing drifts across years. Let $\mathbf{W} \in \mathbb{R}^{N_p \times d}$ denote N_p learned period prototypes. We obtain period

assignment scores and aggregate periodic features as

$$\alpha^{(\ell)} = \text{softmax}\left(\mathbf{S}^{(\ell)}\mathbf{W}^\top\right), \quad \mathbf{V}^{(\ell)} = \alpha^{(\ell)}\mathbf{W}, \quad (7)$$

where $\alpha^{(\ell)} \in \mathbb{R}^{(T+H) \times N_p}$ and $\mathbf{V}^{(\ell)} \in \mathbb{R}^{(T+H) \times d}$. The final period representation is

$$\mathbf{H}_P^{(\ell)} = \text{LN}\left(\mathcal{W}_P \mathbf{V}^{(\ell)} + \mathbf{U}^{(\ell)}\right). \quad (8)$$

This branch adds frequency-aware structure while keeping the model small.

3.4. Fusion and Forecasting

The default EPIC block fuses the temporal and period branches additively,

$$\mathbf{G}^{(\ell)} = \mathbf{H}_T^{(\ell)} + \mathbf{H}_P^{(\ell)}. \quad (9)$$

The block output is then formed by a residual update,

$$\bar{\mathbf{Z}}^{(\ell)} = \text{LN}\left(\mathbf{G}^{(\ell)} + \bar{\mathbf{Z}}^{(\ell-1)}\right). \quad (10)$$

Section A compares this default fusion with two single-branch variants that retain only the temporal feature $\mathbf{H}_T^{(\ell)}$ or only the periodic feature $\mathbf{H}_P^{(\ell)}$. Because $\mathbf{H}_P^{(\ell)}$ already includes the temporal residual $\mathbf{U}^{(\ell)}$, the latter variant is a period-enhanced branch rather than a pure frequency-only model.

After the final block, a linear readout maps latent states back to the target space,

$$\hat{\mathbf{X}} = \mathcal{W}_{\text{out}}\left(\bar{\mathbf{Z}}^{(L)}\right), \quad (11)$$

and the forecast is taken from the last H steps,

$$\hat{\mathbf{Y}} = \hat{\mathbf{X}}_{T+1:T+H}. \quad (12)$$

Finally, the predictions are de-normalized using the instance statistics of the input sequence. The overall architecture is compact and explicitly period-aware, without a general-purpose temporal attention backbone.

4. Experiments

4.1. Experimental Setup

We evaluate on the standard weekly `national_illness` benchmark used in multivariate long-horizon forecasting. The dataset has seven channels. We use a look-back window of 36 and horizons $H \in \{24, 36, 48, 60\}$. We report MSE and MAE at each horizon and their four-horizon average. Unless otherwise stated, EPIC denotes the reference configuration averaged over three random seeds. In all experiments we use $L = 2$ forecasting blocks, latent dimension $d = 128$,

and $N_p = 5$ learnable period prototypes. Table 1 makes the benchmark setup explicit. Across the full retrospective panel we observe 966 weekly rows, split into 676/97/193 train/validation/test rows. Under the fixed 36-step look-back, carrying boundary context expands the validation and test segments to 133 and 229 effective rows, yielding 74/62/50/38 validation windows and 170/158/146/134 test windows for $H = 24/36/48/60$, respectively.

Table 1. Split sizes and horizon-specific sliding-window counts on `national_illness` under the fixed 36-step look-back and 18-step label context. ‘Effective rows’ count the context-extended validation and test segments after carrying the preceding look-back history across split boundaries.

Split	Rows	Effective rows	W24	W36	W48	W60
TRAIN	676	676	617	605	593	581
VAL	97	133	74	62	50	38
TEST	193	229	170	158	146	134
TOTAL	966	1038	861	825	789	753

The full-history, ACF, and spectral panels of Figure 1 confirm three features of the target series: short-lag persistence, annual periodicity, and an exception in the 2007–2010 interval coinciding with the 2009 H1N1 pandemic¹.

4.2. Baselines on the Benchmark

Our comparison set on the benchmark includes standard supervised forecasters such as PatchTST (Nie et al., 2023), WhiteTST (Shen et al., 2024), xLSTMTIME (Alharthi & Mahmood, 2024), MSCformer (Li et al., 2024), Autoformer (Wu et al., 2021), Non-stationary Transformer (Liu et al., 2022), and TimesNet (Wu et al., 2023), together with LLM-based or pretrained comparators such as OFA (Zhou et al., 2023), Time-LLM (Jin et al., 2024), UniTime (Liu et al., 2024), TimeCMA (Liu et al., 2025), MIFlu (Moon et al., 2025), and TimesFM (Das et al., 2024).

Tables 2 and 3 compare published benchmark results with EPIC. MSCformer and Time-LLM are nearly tied on Avg MSE, 1.434 versus 1.435, and MIFlu is close at 1.436. This does not show a clear average-error advantage for the LLM-based comparator over the compact baseline. EPIC reaches Avg MSE 1.173, an 18.2% reduction relative to MSCformer and Time-LLM. Table 3 shows a similar margin in MAE. Gains are largest on the four-horizon average and on shorter-to-medium horizons.

Figure 2 complements the aggregate tables with four qualitative examples at horizon $H = 24$ using 36 observed weeks. The four panels are drawn from separated parts of the test period so that they illustrate distinct forecast situations rather than near-duplicates. In these examples, the EPIC forecast is frequently among the closest traces to the

¹[https://www.who.int/emergencies/situations/influenza-a-\(h1n1\)-outbreak](https://www.who.int/emergencies/situations/influenza-a-(h1n1)-outbreak)

Table 2. ILI comparison on the retrospective benchmark in MSE. Lower is better. Published baselines are taken from reported ILI results under the matching benchmark and horizon setting.

Method	H=24	H=36	H=48	H=60	Avg
EPIC	0.818	1.009	1.235	1.630	1.173
MSCformer	1.215	1.382	1.411	1.726	1.434
Time-LLM	1.285	1.404	1.523	1.531	1.435
MIFlu	1.542	1.422	1.414	1.364	1.436
WhiteTST	1.069	1.382	1.461	1.863	1.443
PatchTST	1.319	1.579	1.553	1.470	1.480
Autoformer	3.483	3.103	2.669	2.770	3.006
Non-Stationary	2.294	1.825	2.010	2.178	2.077
TimesNet	2.317	1.972	2.238	2.027	2.139
xLSTMTime	1.514	1.519	1.500	1.418	1.488
TimeCMA	1.996	1.906	1.867	1.920	1.922
OFA	2.063	1.868	1.790	1.979	1.925
UniTime	2.460	1.998	1.979	2.109	2.137

Table 3. ILI comparison on the retrospective benchmark in MAE. Lower is better. Published baselines are taken from reported ILI results under the matching benchmark and horizon setting.

Method	H=24	H=36	H=48	H=60	Avg
EPIC	0.573	0.653	0.710	0.823	0.690
MSCformer	0.728	0.743	0.767	0.835	0.768
Time-LLM	0.727	0.814	0.807	0.854	0.801
MIFlu	0.726	0.779	0.757	0.719	0.745
WhiteTST	0.634	0.732	0.771	0.878	0.753
PatchTST	0.754	0.870	0.815	0.788	0.807
Autoformer	1.287	1.148	1.085	1.125	1.161
Non-Stationary	0.945	0.848	0.900	0.963	0.914
TimesNet	0.934	0.920	0.940	0.928	0.931
xLSTMTime	0.694	0.722	0.725	0.715	0.714
TimeCMA	0.998	0.915	0.868	0.904	0.921
OFA	0.881	0.892	0.884	0.957	0.903
UniTime	0.954	0.912	0.912	0.938	0.929

target, while some comparison traces show larger phase or amplitude error. This qualitative panel complements the benchmark tables with trajectory-level comparisons from the same benchmark setting, together with the two zero-shot TimesFM variants analyzed separately in Appendix B. TimesFM-independent forecasts each variable separately, whereas TimesFM-dependent augments each forecast with the remaining variables as auxiliary inputs.

5. Discussion

Ablation readout. Appendix Tables 4 and 5 show that the full model performs best on the four-horizon average and on most individual horizons. The rows should be read along two axes. The variant column distinguishes the full, temporal-only, and period-enhanced models, and the projection column distinguishes the nonlinear projector from the linear control. The third variant corresponds to the period-enhanced branch defined in Section 3.

Removing the period branch raises Avg MSE from 1.173 to 1.222 (temporal-only), and removing the temporal branch raises it to 1.265 (period-enhanced). The MAE ranking is identical: $0.690 < 0.699 < 0.710$. Temporal-only is marginally lower at $H = 48$, but the full model is lower

on the four-horizon average and on all other horizons. The nonlinear projector also reduces the four-horizon average error across all three variants relative to the linear control.

Scope and caveats. The main exception is the 60-step horizon, where EPIC records MSE 1.630, compared with 1.531 for Time-LLM, 1.470 for PatchTST, and 1.364 for MIFlu. The MAE table tells the same story: EPIC records 0.823, worse than PatchTST (0.788), MIFlu (0.719), and xLSTMTime (0.715). We therefore do not claim uniform superiority at the longest horizon. Accordingly, we limit our claim to average performance on the published `national_illness` benchmark.

The appendix reports a separate zero-shot TimesFM study that includes both a matched $T = 36$ comparison and a longer-context sensitivity analysis.

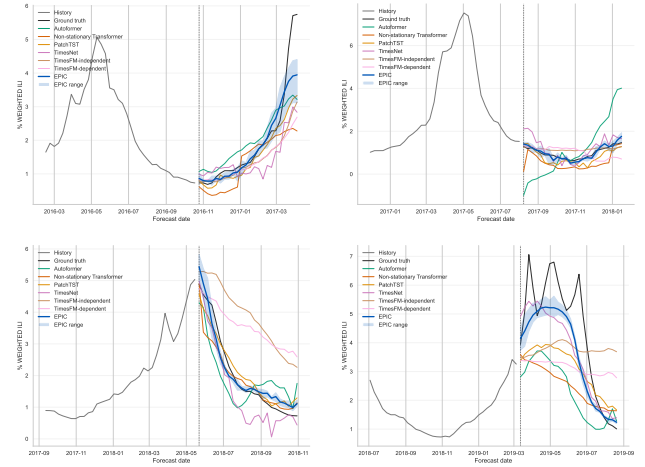


Figure 2. Qualitative forecast comparisons at horizon $H = 24$ using 36 observed weeks. Each panel compares the proposed EPIC configuration with PatchTST, Autoformer, Non-stationary Transformer, TimesNet, and two zero-shot TimesFM variants under the same seven-variable forecasting setting.

6. Conclusion

We presented EPIC, a compact attention-free forecaster for weekly influenza-like illness. On the published `national_illness` benchmark, EPIC improves the four-horizon average over compact and LLM-based comparators, while $H = 60$ remains a limitation.

For weekly ILI forecasting, separating short-range propagation from slowly drifting annual structure improves the four-horizon average compared to a single general-purpose mixer. On this benchmark, a small domain-specific inductive bias outweighs the pretraining scale of LLM-based comparators. Extending the framework to longer horizons, probabilistic forecasting, and additional epidemic datasets remains future work.

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A. EPIC Ablation Details

Tables 4 and 5 report the full ablation results summarized in Section 5. The rows compare three branch variants (full, temporal-only, and period-enhanced) under two projector choices (nonlinear and linear). The period-enhanced variant is the periodic branch from Section 3, applied on top of the temporal feature, so it is not a purely frequency-only model.

Table 4. EPIC ablation on national illness in MSE.

Variant	Projection	H=24	H=36	H=48	H=60	Avg
EPIC	NL	0.818	1.009	1.235	1.630	1.173
Temporal-only	NL	0.849	1.144	1.169	1.725	1.222
Period-only	NL	0.867	1.222	1.267	1.702	1.265
EPIC	L	0.866	1.252	1.240	1.737	1.274
Temporal-only	L	0.936	1.211	1.381	1.765	1.323
Period-only	L	0.993	1.373	1.399	1.686	1.363

Table 5. EPIC ablation on national illness in MAE.

Variant	Projection	H=24	H=36	H=48	H=60	Avg
EPIC	NL	0.573	0.653	0.710	0.823	0.690
Temporal-only	NL	0.583	0.682	0.702	0.830	0.699
Period-only	NL	0.591	0.699	0.720	0.829	0.710
EPIC	L	0.592	0.728	0.731	0.843	0.724
Temporal-only	L	0.626	0.709	0.768	0.849	0.738
Period-only	L	0.629	0.732	0.758	0.828	0.737

The main pattern is stable in both MSE and MAE. With the nonlinear projector, the full model reaches the best four-horizon average, improving Avg MSE from 1.222 and 1.265 for the temporal-only and period-enhanced variants to 1.173, and improving Avg MAE from 0.699 and 0.710 to 0.690. This pattern is consistent with complementary contributions from short-range propagation and recurring periodic structure. The temporal-only model remains slightly better at $H = 48$, but the full model is better on the average and on the other reported horizons.

The projection comparison shows a second consistent pattern. Replacing the nonlinear projector with the linear control increases the average error for all three variants. For the full model, Avg MSE rises from 1.173 to 1.274, and Avg MAE rises from 0.690 to 0.724. The same degradation appears for the temporal-only and period-enhanced rows. Both the joint branch design and the nonlinear projector contribute independently to the four-horizon average error.

B. Matched Zero-Shot TimesFM Comparison

We compare EPIC with zero-shot TimesFM under the same seven-variable forecasting benchmark as in the main pipeline. TimesFM-independent forecasts each channel separately and then stacks the resulting outputs. TimesFM-dependent forecasts one channel at a time while using the other six channels as auxiliary inputs, with future values supplied by the corresponding independent forecasts. Context length 36 matches EPIC, while {54, 72, 90, 108} probe TimesFM context sensitivity.

Table 6. MSE comparison between EPIC and zero-shot TimesFM variants under multiple input context lengths.

Method	Context (T)	H=24	H=36	H=48	H=60	Avg
EPIC	36	0.818	1.009	1.235	1.630	1.173
TimesFM-independent	36	2.940	4.046	4.983	5.181	4.287
	54	2.478	2.781	3.578	3.295	3.033
	72	2.394	2.609	2.601	2.517	2.530
	90	2.021	1.877	1.779	1.763	1.860
	108	1.655	2.084	2.213	2.404	2.089
TimesFM-dependent	36	3.251	4.169	4.970	5.012	4.351
	54	2.783	2.697	3.948	4.112	3.385
	72	2.662	3.088	3.068	3.066	2.971
	90	2.393	2.134	2.188	2.414	2.282
	108	2.044	2.532	2.982	3.254	2.703

Table 7. MAE comparison between EPIC and zero-shot TimesFM variants under multiple input context lengths.

Method	Context (T)	H=24	H=36	H=48	H=60	Avg
EPIC	36	0.573	0.653	0.710	0.823	0.690
TimesFM-independent	36	1.124	1.356	1.522	1.552	1.388
	54	0.984	1.068	1.208	1.170	1.108
	72	0.975	1.057	1.060	1.069	1.041
	90	0.875	0.883	0.885	0.895	0.885
	108	0.827	0.934	0.973	0.990	0.931
TimesFM-dependent	36	1.198	1.414	1.581	1.605	1.450
	54	1.045	1.119	1.337	1.339	1.210
	72	1.068	1.158	1.155	1.155	1.134
	90	0.961	0.948	0.972	0.998	0.970
	108	0.911	1.021	1.111	1.116	1.040

Tables 6 and 7 show that matching the context length is not enough to make zero-shot TimesFM competitive with trained EPIC. Longer contexts help, especially for TimesFM-independent, but the best zero-shot row still remains above EPIC in both Avg MSE and Avg MAE. We treat the longer-context rows as a sensitivity analysis rather than as a direct substitute for the trained EPIC result.

C. Qualitative Forecast Comparisons

Figures 3 to 6 use the same 36-week context and the same four illustrative cases as Figure 2. The comparison set is fixed across panels, and only the forecast horizon changes. Reading across the four horizons shows how the same local

pattern becomes harder to extrapolate as the prediction window grows. These panels therefore help distinguish whether longer-horizon degradation is driven mainly by phase drift, amplitude mismatch, or both.

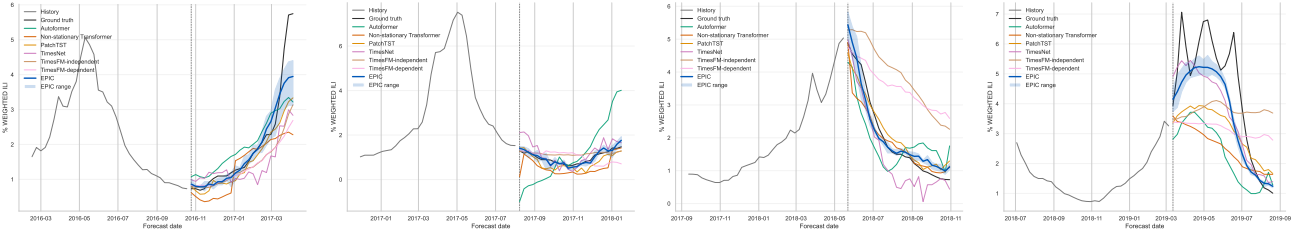


Figure 3. Qualitative forecast comparisons at $H = 24$.

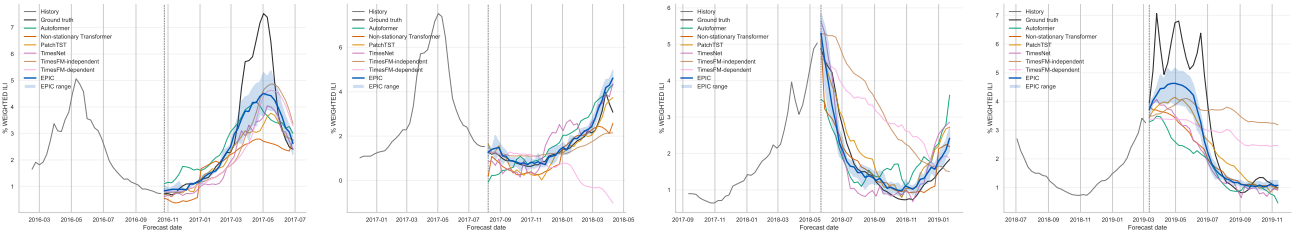


Figure 4. Qualitative forecast comparisons at $H = 36$.

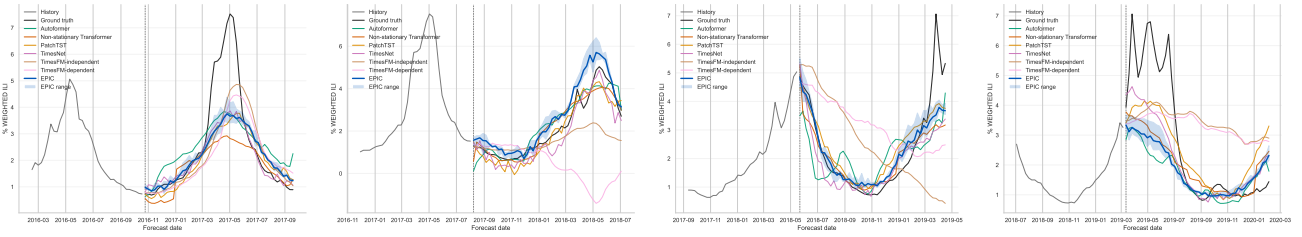


Figure 5. Qualitative forecast comparisons at $H = 48$.

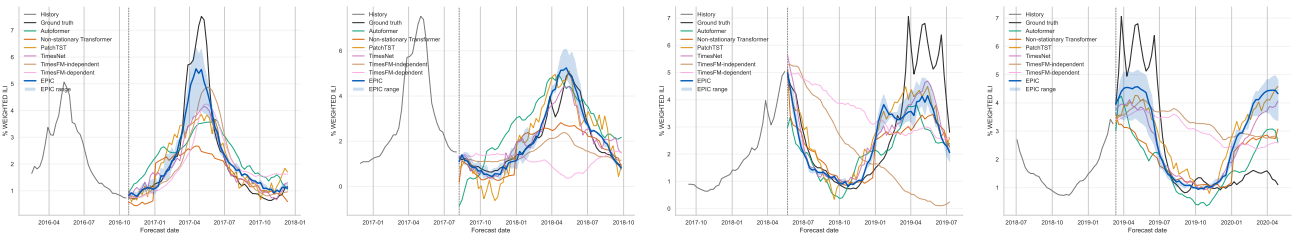


Figure 6. Qualitative forecast comparisons at $H = 60$.